

An Optimality Principle

Let X and Y be sets, and $f : Z \rightarrow \mathbb{R}$, where $Z \subset X \times Y$ and Z is not empty. Consider

$$\bar{V} = \max_{(x,y) \in Z} f(x,y) \quad (1)$$

Assume that this problem has a solution.¹ Next consider

$$V(x) = \max_y f(x,y) \quad \text{subject to} \quad (x,y) \in Z \quad (2)$$

Let $X_0 \subset X$ be the set for which it has a solution (X_0 is not empty since (1) has a solution). Finally consider

$$\max_{x \in X_0} V(x). \quad (3)$$

The next result extends the Composite Commodity Theorem you proved in ASMT I.

Proposition. x^* solves (3) if and only if (x^*, y^*) solves (1) where y^* is any solution to (2) at $x = x^*$. Also $\bar{V} = \max_{x \in X_0} V(x)$.

Proof: Let x^* solve (3). Since $x^* \in X_0$, there is a solution to (2) at $x = x^*$; let y^* be any such solution. I will show that (x^*, y^*) solves (1). Clearly (x^*, y^*) is feasible for (1). Let $(\hat{x}, \hat{y})$ solve (1). Then $\hat{x}$ is in X_0 —otherwise $(\hat{x}, \hat{y})$ could not solve (1). Now

$$\bar{V} = f(\hat{x}, \hat{y}) \leq V(\hat{x}) \leq V(x^*) = f(x^*, y^*)$$

where the first inequality follows from the definition of $V(\cdot)$ and $(\hat{x}, \hat{y}) \in Z$, the second follows since x^* solves (3) and $\hat{x} \in X_0$, and the last equality follows since y^* solves (2) at x^* . The inequality $\bar{V} \leq f(x^*, y^*)$ implies that (x^*, y^*) solves (1).

Now let y^* be any solution to (2) at $x = x^*$ and let (x^*, y^*) solve (1). I will show that x^* solves (3). Again, $x^* \in X_0$, otherwise (x^*, y^*) could not solve (1). Let $\phi(x)$ be any solution to (2) for any $x \in X_0$ with the restriction that $\phi(x^*) = y^*$. Then $V(x) = f(x, \phi(x))$ for all $x \in X_0$. But for any $x \in X_0$, $V(x^*) = f(x^*, y^*) \geq f(x, \phi(x)) = V(x)$, where the inequality follows since (x^*, y^*) solves (1) and $(x, \phi(x)) \in Z$; so x^* solves (3). That $\bar{V} = \max_{x \in X_0} V(x)$ now follows immediately. $\square$

Application to the Composite Commodity Theorem. For the composite commodity theorem interpret y as a vector of goods to be aggregated. For that special case the point $\phi(x)$ is a demand for the y vector facing prices q^0 and income $w - p \cdot x$. The vector x affects the demand for y in two ways: by changing the conditional preferences for y ; and by affecting how much income there is to spend on y . The conditional demand can be written as $d_y(x, w - p \cdot x)$, so $V(\cdot)$ can in this case be written $V(x) = f(x, y') := \tilde{u}(x, w - p \cdot x)$ for any $y' \in d_y(x, w - p \cdot x)$ and some function $\tilde{u}$. The number $w - p \cdot x$ can be interpreted as the composite good z in the problem. (In this discussion I set $\alpha = 1$ in the problem statement.)

¹For example, Z is a nonempty compact subset of a Euclidean space and f is continuous.